

CryptoHedge

Multi-agent AI cryptocurrency hedge fund · delivered MVP

- Capital \$10M · 90-day horizon · single seed = 90909090 (reproducible)
- 11 autonomous AI agents: signal generation, hedging, risk, portfolio
- Δ -v-neutral BTC hedging via the Heston model + diversified portfolio
- MVP delivered and validated on historical backtest (walk-forward + stress tests)
- 42 automated tests · rebalanced backtest · RU/EN dashboards

+42.1%

Return

1.63

Diversification

13.7/15

Effective N

0.47

Confidence

Trading universe & data

Instruments, sources, fees, scaling

- Universe: 100 spot USDT pairs; primary BTCUSDT; BTC options (call/put, 30d, 11 strikes)
- Sources: bundled (offline) · synthetic (generator) · binance (live REST) — config-selected
- Fees: spot 0.03%; Deribit-style options 0.03% + 12.5% price cap
- Hedge: spot for Δ , option for v — neutralise market greeks
- **Why: the MarketDataProvider abstraction isolates exchange APIs → scale to OKX/Bybit/Deribit without rewriting agents**

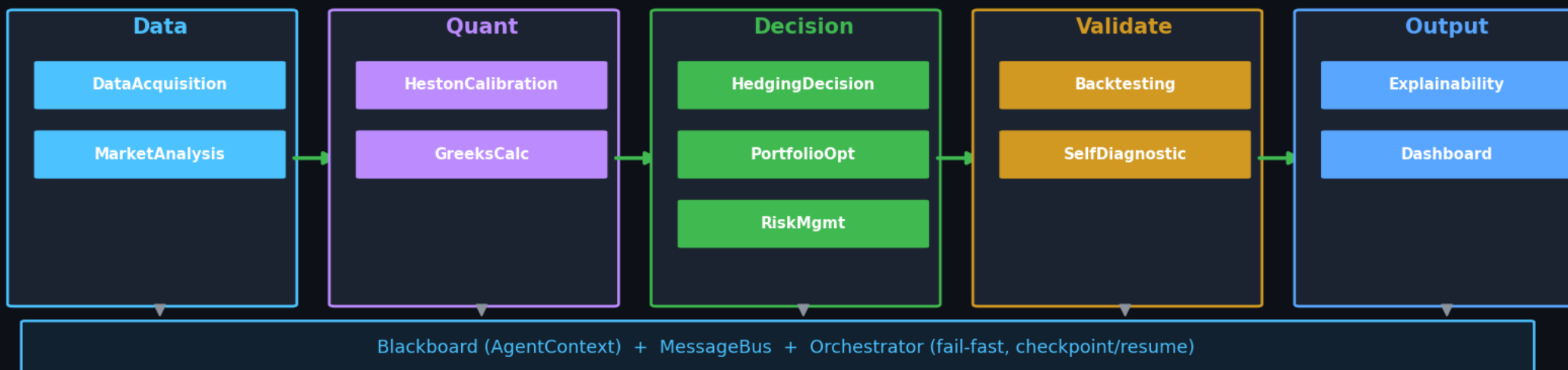


Swap provider → OKX / Bybit / Deribit without touching agents

Multi-agent architecture

Loose coupling, traceability, testability

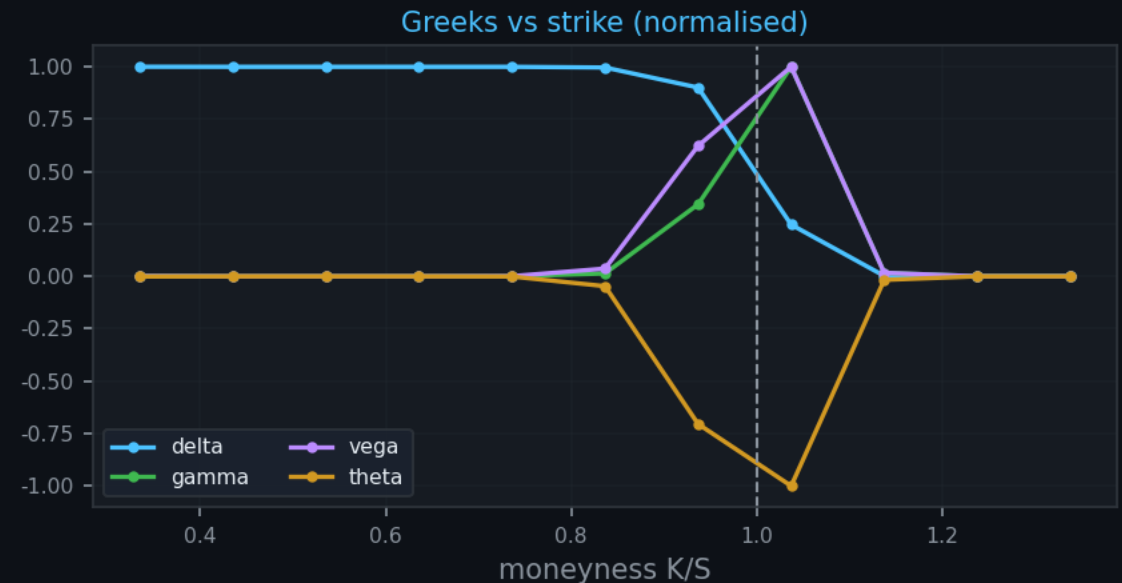
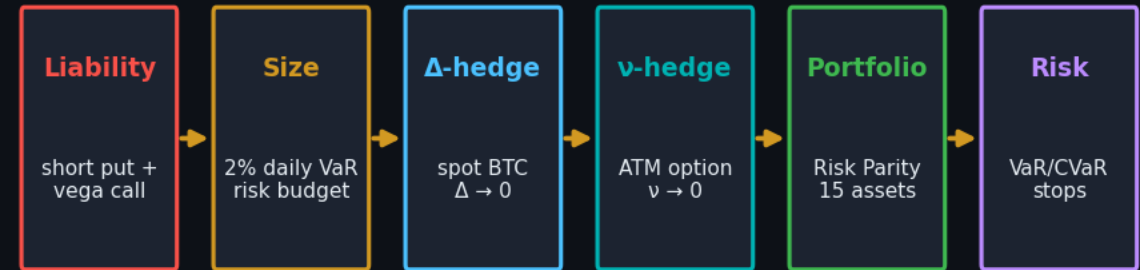
- Blackboard (AgentContext): agents exchange artifacts with no direct calls
- MessageBus: typed messages with correlation_id — full decision traceability
- Clean Architecture: agents → services → domain → pyquant
- Fail-fast orchestrator; checkpoint/resume — robust long runs
- **Why blackboard: adding/replacing an agent never breaks others — extensibility and isolated tests**



Quantitative engine & hedging

Section 7: portfolio greeks · Heston model

- Heston calibration (MLE + IV surface); Black-Scholes / SABR benchmarks
- Full greeks set: Δ Γ ν Θ ρ + vanna / volga / charm
- Liability book: short put (0.95 K/S) + vega call; hedge size from 2% daily VaR budget
- Δ hedge via spot, ν hedge via option $\rightarrow$ residual greeks ≈ 0
- **Why Heston: stochastic volatility beats BS on crypto** (smile, vol clustering)

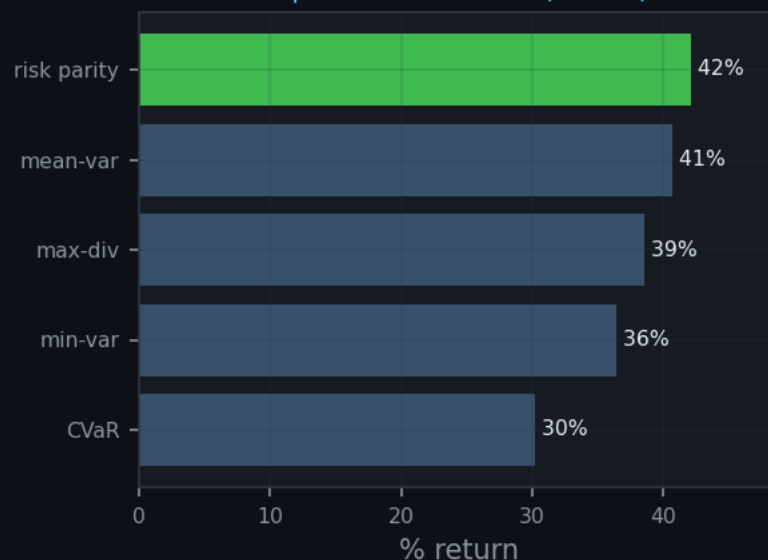


Quantitative portfolio management

Section 9: optimization, rebalancing, diversification

- 5 optimizers: Mean-Variance · Risk Parity · Min-Var · Max-Div · CVaR
- Auto-selected via rebalanced backtest (fees included)
- MVP: selected risk parity → return +42.1%, DR 1.63, effective N 13.7/15
- Rebalance every 5 days; max weight 20% — concentration control
- **Why Risk Parity: robust diversification without relying on return forecasts**

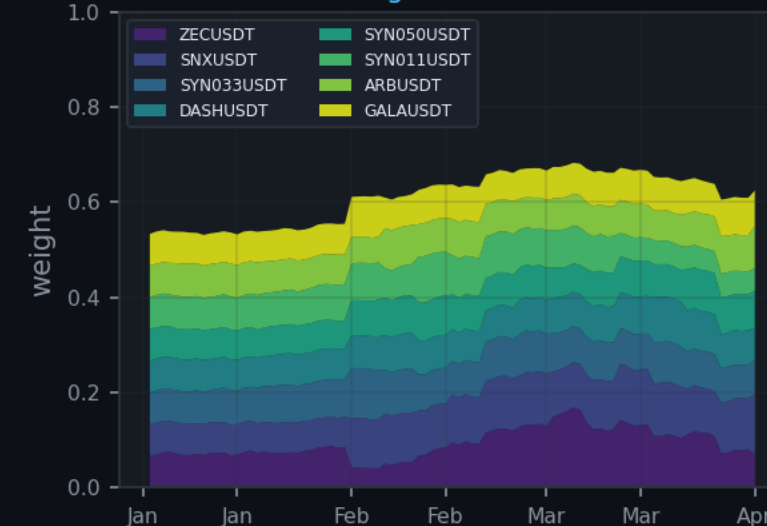
Optimizer backtest (return)



Portfolio equity & rebalances



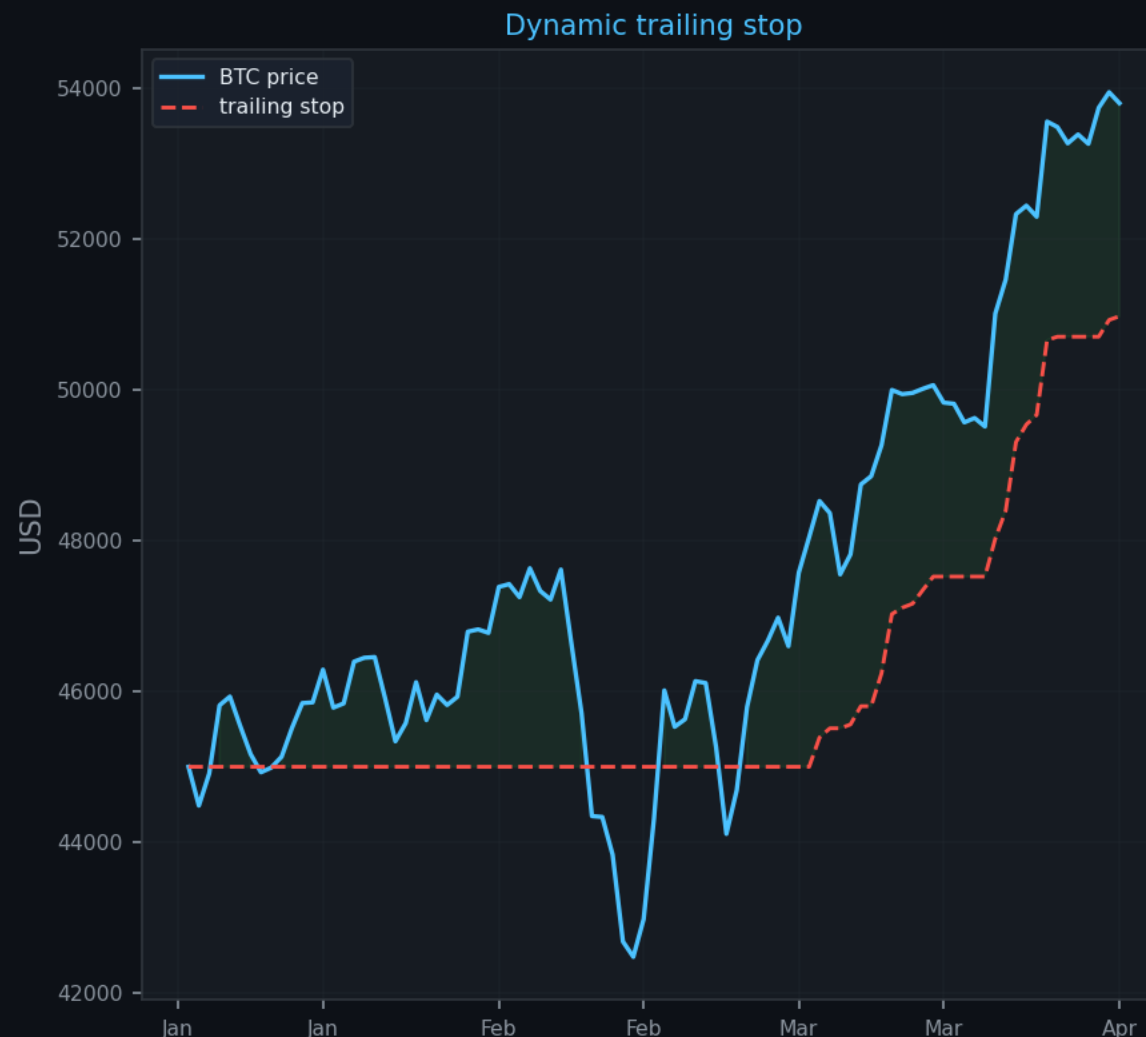
Weight evolution



Risk management

Section 10: VaR / CVaR and adaptive stops

- Risk measures: VaR (hist/Gauss/Cornish-Fisher), CVaR/ES, max drawdown — VaR 0.0% $\ll$ 5% limit
- Adaptive stop-loss: $\max(\text{ATR}, \text{VaR}, \text{Heston-vol})$ — reacts to volatility regime
- Dynamic trailing stop; delta-balance zones (green / red)
- Protected: market Δ , vega, concentration, tail shocks
- Not protected: execution/slippage, counterparty, basis, jumps
- **Assumptions: daily bars, long-only, historical VaR, modelled fees**



Protective mechanisms

Layered portfolio defence

- Trailing stop ($\text{ATR} \times 2.5$) — lock in profit on reversal
- Dynamic hedge adjustment — rebalance Δ/v on greek drift
- Position sizing — from risk budget, not from a market view
- Exposure limits — max weight, 3× leverage, 25% drawdown cap
- Portfolio rebalancing — return to target weights every 5 days

Position sizing

from risk budget, not market view

Exposure limits

max weight 20% · leverage 3× · MDD 25%

Dynamic hedge

rebalance Δ / v on greek drift

Adaptive + trailing stop

$\max(\text{ATR}, \text{VaR}, \text{Heston-vol})$

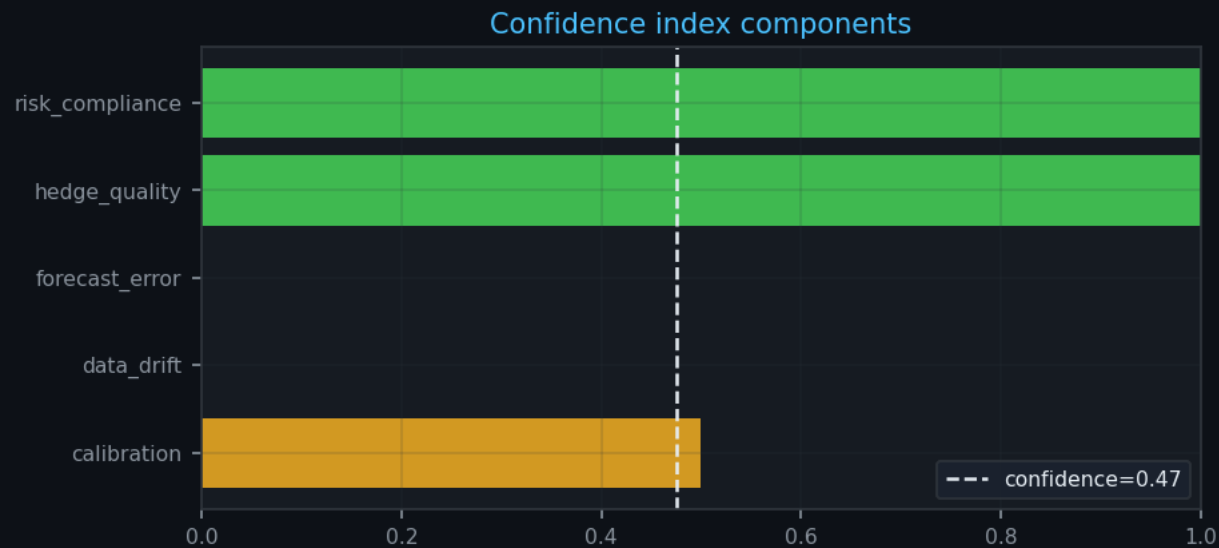
Portfolio rebalancing

back to target weights every 5d

Validation, backtest & self-diagnostics

Sections 11-12: stress tests and confidence index

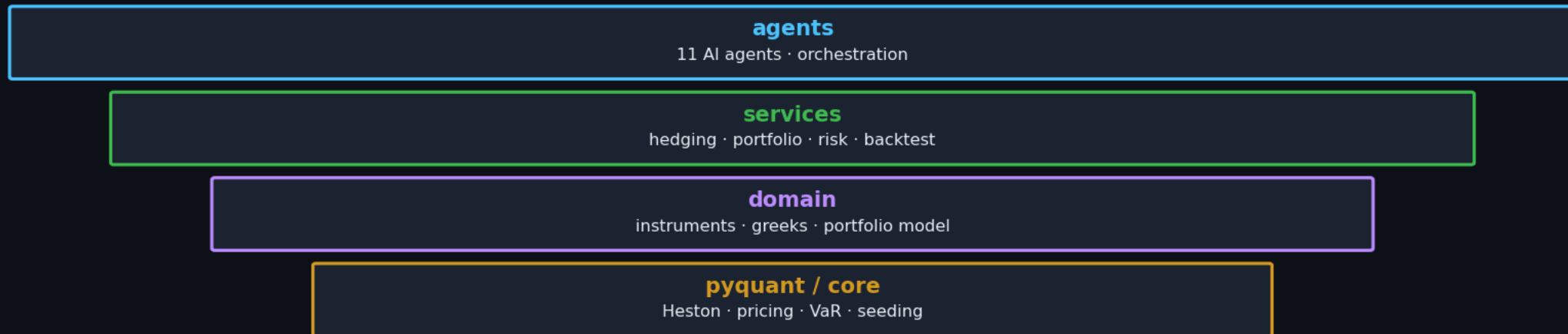
- Walk-forward: train 30d / test 5d; look-ahead and survivorship-bias controls
- Stress tests: hedged book vs naked BTC
- Self-Diagnostic: confidence index (calibration, data-drift PSI, forecast-error, hedge-quality, risk-compliance)
- Current confidence = 0.47: under data drift the system honestly flags low_confidence
- **MVP passed historical backtesting; results are reproducible**



Reproducibility & engineering

Determinism, tests, clean architecture

- Single seed (90909090) fixes all RNG: data, calibration, PnL, weights
- 42 automated tests; end-to-end solution.ipynb; $\max|\Delta\text{PnL}| = 0$ on re-run
- Checkpoint/resume; config-driven pipeline (YAML + Pydantic), no hardcoding
- Clean Architecture: business logic independent of data providers and exchanges
- **Why: reproducibility is mandatory for audit and fair strategy comparison**



Dependencies point inward → business logic independent of exchanges/data

Future development

Roadmap

- Fully automated trade execution across multiple CEX (live)
- Telegram bot for control and monitoring
- News feeds and sentiment analysis integration
- Additional data sources; reinforcement-learning agents
- Cross-exchange arbitrage · live options market · self-adapting agent coordination

